

Submission MICRO1-PS1-SUB01

Question MICRO1-PS1-Q01

scratch: show step

1. $33 - 5P = 3 + P \Rightarrow P^* = (33 - 3) / (5 + 1) = 5.00$
 2. $Q^* = 33 - 5(5.00) = 8.00$
 3. substitution gives $Q_d = Q_s$
- => FINAL: $P^* = 5.00$, $Q^* = 8.00$ (check units/interpretation)

Question MICRO1-PS1-Q02

scratch: show step

1. $\% \Delta Q = -2 / 19.0$
 2. $\% \Delta P = 0.0952$
 3. $E = -1.11$; $|E| > 1$ (elastic)
- => FINAL: elasticity = -1.11; elastic (check units/interpretation)

Question MICRO1-PS1-Q03

scratch: show step

1. Shift the wedge between buyer and seller prices by t.
 2. Quantity falls relative to equilibrium.
 3. The lost mutually beneficial trades create a positive deadweight-loss triangle.
- => FINAL: buyer price minus seller price equals tax; quantity falls; deadweight loss increases (check units/interpretation)

Question MICRO1-PS1-Q04

scratch: show step

1. $SMC = PMC + \text{marginal external cost}$, so SMC lies above PMC.
 2. A Pigouvian tax equal to marginal external cost aligns private and social incentives.
- => FINAL: social MC above private MC; Pigouvian tax can internalize the external cost (check units/interpretation)

Question MICRO1-PS1-Q05

scratch: show step

1. At the ceiling, Q_d exceeds Q_s .
 2. Shortage = $Q_d - Q_s$; queues or rationing allocate the scarce quantity.
- => FINAL: $Q_d > Q_s$ creates a shortage; queues/rationing may allocate units (check units/interpretation)

Question MICRO1-PS1-Q06

scratch: show step

1. $35 - 4P = 3 + P \Rightarrow P^* = (35 - 3) / (4 + 1) = 6.40$
 2. $Q^* = 35 - 4(6.40) = 9.40$
 3. substitution gives $Q_d = Q_s$
- => FINAL: $P^* = 6.40$, $Q^* = 9.40$ (check units/interpretation)